

PC-03 · EMULSIFIERS & THICKENERS · PERSONAL CARE

HS GMS 95

Glyceryl Stearate

INCI Glyceryl Stearate	CAS 31566-31-1	EC —	FAMILY PC-03	SUPPLIED AS Flakes / beads
---------------------------	-------------------	---------	-----------------	-------------------------------

01 POSITIONING

The invisible architecture of every cream on the shelf.

02 HOW IT WORKS

- 01 Crystalline scaffold

Fatty chains crystallize into a lamellar network that gives creams their body and stability.
- 02 Co-emulsification

GMS reinforces the interfacial film and narrows droplet distribution.
- 03 Melting-point design

Stearic-based systems set sticks and balms at skin-melt temperatures.

03 APPLICATIONS

- Creams and lotions
- Sticks, balms and deodorants
- Hair conditioners (body)
- Soap superfatting
- Candles and technical uses

04 FORMULATION GUIDE

USE LEVEL	1–8%
PH & CONDITIONS	Stable pH 4–9 in emulsified systems
— Melt into the oil phase at 70–75 °C; homogenize before cooling through the crystallization window.	
— Combine GMS with stearic acid for the classic self-bodying pair.	

05 TYPICAL SPECIFICATION

Appearance	White flakes / beads
Purity / monoester	per grade (GMS ≥ 95% mono)
Melting point	54–70 °C
Acid value	per TDS

Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE – DIRECTIONAL

Emulsion stability (cycle test, class data)
+ 2–4% fatty base

Body/viscosity per dosage
Fatty-base route

Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.